

DHARITREE BEZBORUAH

PERSONAL INFORMATION

Date of Birth : 17 October, 1999
Nationality : Indian
Pronoun: She/Her
Present Address : Department of Physics
Tezpur University
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Languages : Assamese, English, Hindi

EDUCATION

- *PhD in Physics* September, 2022 - Present
Tezpur University, Assam, India
Guide: Dr. Moon Moon Devi.
Current area of research:
1. Neutrino phenomenology,
2. Use of Machine Learning (ML) in high energy physics.
- *Masters of Science in Physics* 2020 - 2022
Tezpur University, Assam, India
CGPA: 8.52 (85.2 %)
- *Bachelor of Science in Physics* 2017 - 2020
Cotton University, Assam, India
CGPA: 8.44 (84.4 %)
- *HSSLC* 2015 - 2017
Salt Brook Academy – Assam, India
Percentage: 93
Physics, Chemistry, Mathematics, Biology
- *HSLC* 2015
Sankardev Sishu Niketon, Dhakuakhana – Assam, India
Percentage: 92.12

PAST RESEARCH EXPERIENCES

- *Exploring non-standard effects in long-baseline neutrino experiments.* 2021 - 2022
Masters project in Tezpur University
Mentored by Dr. Moon Moon Devi, Assistant Professor, Department of Physics
- *Mukhanov Sasaki Equation* June 2021 - October 2021
IAS-INSa-NASI summer research Fellow at Indian Institute of Astrophysics.
Mentored by Prof. Pravabati Chingangbam

ACHIEVEMENTS

- Best Poster Award Honorable mention at XXVI DAE-BRNS High Energy Physics Symposium 2024, Benaras Hindu University.

- CSIR-NET (LS) 2023.
- Innovation in Science Pursuit for Inspired Research (INSPIRE)- Fellowship for Ph.D., January 2023- Present.
- State Eligibility Test North East Region in 2023.
- Innovation in Science Pursuit for Inspired Research (INSPIRE)-Scholarship for higher education(SHE) by Department of Science and Technology (DST), India from 2017-2022.
- Selected as IAS-INSA-NASI Summer Research Fellow, 2021.

TECHNICAL SKILLS

- Programming languages: Python, C, C++
- Machine Learning: Scikit-learn, PyTorch, TensorFlow
- Operating systems: Linux, Windows
- Software: LaTeX, ROOT, Mathematica, GLoBES, GNUPLOT, basic MATLAB

PUBLICATIONS

Peer Reviewed Articles

- **D. Bezboruah**, A. Medhi, M. M. Devi, “*Dark NSI & neutrino oscillations : probing via δ_{CP} measurements at DUNE and T2HK*”, [arXiv: 2601.04802] (Under Review).
- A. Sarker, **D. Bezboruah**, A. Medhi, M. M. Devi, “*Sensitivity of DUNE in Presence of Off-Diagonal Scalar NSI Parameters*”, Phys.Rev.D 112 (2025) 3, 035042 [arXiv: 2406.15307].
- **D. Bezboruah**, D. S. Chattopadhyay, A. Medhi, A. Sarker, M. M. Devi, “*Neutrino oscillations in presence of diagonal elements of scalar NSI: an analytic approach*”, JHEP 12 (2025) 222 [arXiv:2410.05250].
- A. Sarker, A. Medhi, **D. Bezboruah**, M. M. Devi, D. Dutta, “*Impact of scalar NSI on the neutrino mass hierarchy sensitivity at DUNE, HK, and KNO*”, JHEP 06 (2024) 128 [arXiv:2309.12249].

Conference Proceedings

- **D. Bezboruah**, A. Medhi, M. M. Devi, “*Dark Non-standard Interaction and CP-Phase Measurement at DUNE*”, Springer Proc. Phys. 432 (2026) 1244-1247.
- **D. Bezboruah**, D. S. Chattopadhyay, A. Medhi, A. Sarker, M. M. Devi, “*Effect of Scalar NSI on Neutrino Oscillations: An Analytic Approach*”, Springer Proc. Phys. 432 (2026) 1248-1251.
- A. Sarker, A. Medhi, **D. Bezboruah**, D. Dutta, M. M. Devi, “*Scalar NSI and Neutrino Mass Ordering Sensitivities in Long-Baseline ν -Experiments*”, Springer Proc. Phys. 432 (2026) 1240-1243.
- A. Sarker, **D. Bezboruah**, A. Medhi, M. M. Devi, “*Probing CP-Violation Sensitivities at DUNE in Presence of Off-Diagonal Scalar NSI*”, Springer Proc. Phys. 432 (2026) 1252-1255.
- A. Medhi, A. Sarker, **D. Bezboruah**, M. M. Devi and D. Dutta, “*Scalar NSI: A Unique Probe to Explore New Physics in Neutrinos*”, Springer Proc. Phys. **361** (2025), 29-41, doi:10.1007/978-981-97-7441-8_4.
- M. M. Devi, **D. Bezboruah**, A. Medhi, A. Sarker and D. Dutta, “*Neutrino mass ordering sensitivities at DUNE, HK and KNO in presence of scalar NSI*”, Contribution to NuFact2024 [arXiv:2412.20026].
- M. M. Devi, A. Medhi, A. Sarker, **D. Bezboruah**, “*Imprints of scalar mediated NSI on long baseline experiments*”, Zenodo, 2023, <https://doi.org/10.5281/zenodo.8347338>.

- **D. Bezboruah** [On behalf of GRAPES-3 Collaboration], “*Machine learning model for separation of muons from punch-through hadrons in EAS at GRAPES-3 experiment*”, PoS ICRC2023 (2023) 522.
- **D. Bezboruah**, A. Medhi, and M. M. Devi, “*The effect of dark non-standard interactions on the CP-phase measurement at DUNE*”, PoS ICRC2023 (2023) 1201.
- A. Medhi, **D. Bezboruah**, D. Dutta, and M. M. Devi, “*Probing scalar Non-Standard Interactions at DUNE, T2HK and T2HKK*”, PoS ICHEP2022 590.

TALKS AND PRESENTATIONS

- “Analytic Approximation of Neutrino Oscillations in Presence of Scalar NSI” at TU-GradSciCon-2025.
- “Effect of scalar NSI on neutrino oscillations: An Analytic Approach (Poster)” at XXVI DAE-BRNS High Energy Physics Symposium 2024. [*Best Poster, Honorable Mention*].
- “Dark Non-Standard Interaction and CP-phase measurement at DUNE (Poster)” at XXVI DAE-BRNS High Energy Physics Symposium 2024.
- “Impact of Scalar NSI on Neutrino Oscillations: An Analytic Approach” at PANE-2024.
- “Probing Non-Standard Interactions and CP-violation at DUNE via ν -oscillations (Poster)” at NEMA-X.
- “Machine learning model for separation of muons from punch-through hadrons in EAS at GRAPES-3 experiment (Poster)” at ICRC 2023.
- “The effect of dark non-standard interactions on the CP-phase measurement at DUNE (Poster)” at ICRC 2023.

CONFERENCES, SCHOOLS AND WORKSHOPS

- IIFC- ν P (Indian-Institutions Fermilab Collaboration in Neutrino Physics) School 2025 at NISER, Bhubaneswar, in October 2025.
- School on Experimental Astroparticle Physics 2025 at Cosmic Ray Laboratory - TIFR, Ooty, in March 2025.
- XXVI DAE-BRNS High Energy Physics Symposium 2024 at Benaras Hindu University in December 2024.
- XIV Biennial Conference of Physics Academy of the North-East at Tezpur University on November 2024.
- North-East Meet of Astronomers (NEMA) - X, Tezpur University in October 2024.
- Understanding the Universe Through Neutrinos at ICTS Bangalore in April 2024
- Experimental High Energy Particle Physics School on software development, TIFR, Mumbai in January 2024.
- Statistical Methods and Machine Learning in High Energy Physics, ICTS Bangalore in September 2023.
- 38th International Cosmic Ray Conference (ICRC2023) held at Nagoya University in Nagoya, Japan (Online) in July 2023.
- ICFA school on instrumentation in high energy physics, TIFR Mumbai in February 2023.
- Online participation in Niels Bohr Institute Neutrino Summer School in July 2022.

- Indian Institute of Astrophysics online Summer School in July 2021.

VISITS

- Group of Prof. Srubabati Goswami, Theoretical Physics Division, Physical Research Laboratory (PRL), Ahmadabad, India, February 5 - February 11, 2026.
- Group of Prof. Srubabati Goswami, Theoretical Physics Division, Physical Research Laboratory (PRL), Ahmadabad, India, January 7 - January 28, 2025.
- Visit to Cosmic Ray Laboratory (CRL), Ooty, India, January 2023.

REFEREES

Available if required.

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